

Homework 3: Data Warehousing for Data Engineering Zoomcamp 2026

Due date: 9 February 2026 20:59 (local time)

Questions

Question 1. What is count of records for the 2024 Yellow Taxi Data? (1 point)

- ☐ 65,623
☐ 840,402
☒ 20,332,093
☐ 85,431,289

Question 2. What is the estimated amount of data that will be read when this query is executed on the External Table and the Table? (1 point)

- ☐ 18.82 MB for the External Table and 47.60 MB for the Materialized Table
☒ 0 MB for the External Table and 155.12 MB for the Materialized Table
☐ 2.14 GB for the External Table and 0MB for the Materialized Table
☐ 0 MB for the External Table and 0MB for the Materialized Table

Question 3. Why are the estimated number of Bytes different? (1 point)

- ☐ BigQuery is a columnar database, and it only scans the specific columns requested in the query. Querying two columns (PULocationID, DOLocationID) requires reading more data than querying one column (PULocationID), leading to a higher estimated number of bytes processed.
☐ BigQuery duplicates data across multiple storage partitions, so selecting two columns instead of one requires scanning the table twice, doubling the estimated bytes processed.
☐ BigQuery automatically caches the first queried column, so adding a second column increases processing time but does not affect the estimated bytes scanned.
☒ When selecting multiple columns, BigQuery performs an implicit join operation between them, increasing the estimated bytes processed

Question 4. How many records have a fare_amount of 0? (1 point)

- ☐ 128,210
☐ 546,578
☐ 20,188,016
☒ 8,333

Question 5. What is the best strategy to make an optimized table in Big Query if your query will always filter based on tpep_dropoff_datetime and order the results by VendorID (Create a new table with this strategy) (1 point)

- ☒ Partition by tpep_dropoff_datetime and Cluster on VendorID
☐ Cluster on by tpep_dropoff_datetime and Cluster on VendorID
☐ Cluster on tpep_dropoff_datetime Partition by VendorID
☐ Partition by tpep_dropoff_datetime and Partition by VendorID

Question 6. Write a query to retrieve the distinct VendorIDs between tpep_dropoff_datetime 2024-03-01 and 2024-03-15 (inclusive). Use the materialized table you created earlier in your from clause and note the estimated bytes. Now change the table in the from clause to the partitioned table you created for question 5 and note the estimated bytes processed. What are these values? (1 point)

- ☐ 12.47 MB for non-partitioned table and 326.42 MB for the partitioned table
☒ 310.24 MB for non-partitioned table and 26.84 MB for the partitioned table
☐ 5.87 MB for non-partitioned table and 0 MB for the partitioned table
☐ 310.31 MB for non-partitioned table and 285.64 MB for the partitioned table

Question 7. Where is the data stored in the External Table you created? (1 point)

- ☐ Big Query
☐ Container Registry
☒ GCP Bucket
☐ Big Table

Question 8. It is best practice in Big Query to always cluster your data: (1 point)

- ☐ True
☒ False

Question 9. Write a 'SELECT count(*)' query FROM the materialized table you created. How many bytes does it estimate will be read? Why? (not graded)

Estimated Bytes: 0 B

Reasoning: BigQuery maintains metadata (storage statistics) for all native materialized tables. This metadata includes the total row count for the table. When a query performs a naked SELECT count(*) without any filters, the execution engine retrieves this value

Homework URL

<https://github.com/dammi01/data-engineering-zoomcamp>

Learning in public links (optional)

https://www.linkedin.com/posts/michael-dambeck_dataengineeringzoomcamp-dataengineeri

<https://x.com/mdambeck/status/2020978810901544913?s=20>

Add Link

Time spent on lectures (hours) (optional)

4

Time spent on homework (hours) (optional)

5

FAQ contribution (FAQ document, optional)

stored in **Google Cloud**. BigQuery may report 0 bytes in the estimator but will still consume about milliseconds as it has to read file footers/headers to calculate the count, since it doesn't own the metadata layer for external files.

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Status: Not submitted